Optimality Thresholds in Directional Cost: When Does the Diagonal Cease to Be Optimal?

Pairing, Cyclic, and Full-Connection Costs on S_+^{n-1}

Kaifan Xie

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Abstract

Fix a direction-dependent cost $d(e, \lambda) = A(e)(1 + \lambda C(e))$ on the positive unit sphere S_+^{n-1}, where $A(e) = n/\sum_i e_i$ and C is a normalised directional penalty. The diagonal direction e_{diag} achieves cost $d_{\text{diag}}(\lambda) = \sqrt{n}(1 + \lambda)$. We determine, for three natural cost families, the smallest λ_0 at which a non-diagonal direction first matches the diagonal—the *optimality threshold*.

For *pairing costs* in even dimensions $n = 2k$, the threshold $\lambda_0 = \frac{1}{3}$ is independent of k, arising from the double root of $-2k(3\lambda - 1)^2 = 0$. For *cyclic costs* in odd dimensions, $\lambda_0(n) = 1/(1 + 2\cos(\pi/n))$, proved via Fourier decomposition of the cyclic cost matrix. The case $n = 3$ is characterised by the perfect-square identity $(s - \sqrt{3})^2 \geq 0$; the bifurcation from $n = 3$ to $n \geq 5$ is marked by the factor $n - 3$ in an exact asymptotic formula. For the *full-connection cost* in any dimension n, $\lambda_0 = (n - 1)/(n + 1)$, with the equality set being the full $(n - 2)$-sphere and the cost corresponding to the A_{n-1} root lattice. Asymptotically, all three families satisfy $\lambda_0 \to \frac{1}{3}$ as $n \to \infty$ (pairing at rate 0; cyclic at rate $\pi^2/(9n^2)$; full-connection at rate $2/n$ towards 1).

We also characterise the asymmetric-target case $r = (r_1, \ldots, r_n)$ with $r_i > 0$: the threshold $\lambda_0 > 0$ if and only if $r_1 = \cdots = r_n$, with the direction $e^* \propto (1/r_1, \ldots, 1/r_n)$ strictly beating the diagonal for all $\lambda \geq 0$ whenever the r_i are not all equal. This follows from a single application of Cauchy–Schwarz.

All results are proved without numerical input. The threshold values are determined by the algebraic structure of the cost, not solely by the ambient Euclidean geometry.

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1 Setup and Definitions

Definition 1.1 (Directional cost and switch value)**.** Let $e \in S_+^{n-1} := \{e \in \mathbb{R}^n : e_i > 0,\ |e| = 1\}$ and let $C : S_+^{n-1} \to (0, 1]$ be a *normalised directional cost* satisfying $C(e_{\text{diag}}) = 1$ where $e_{\text{diag}} = (1, \ldots, 1)/\sqrt{n}$. For $\lambda \geq 0$, define

$$d(e, \lambda) = A(e)\big[1 + \lambda C(e)\big], \qquad A(e) = \frac{n}{\sum_{i=1}^{n} e_i}.$$

The *diagonal cost* is $d_{\text{diag}}(\lambda) = \sqrt{n}(1 + \lambda)$. For $e \neq e_{\text{diag}}$ with $\sqrt{n} - A(e)C(e) > 0$, the *switch value* is

$$\lambda_{sw}(e) = \frac{A(e) - \sqrt{n}}{\sqrt{n} - A(e)C(e)}.$$

The *optimality threshold* is

$$\lambda_0 = \inf\{\lambda \geq 0 : \exists\, e \neq e_{\text{diag}},\ d(e, \lambda) \leq d_{\text{diag}}(\lambda)\}.$$

Remark 1.2. The diagonal direction satisfies $A(e_{\text{diag}}) = \sqrt{n}$ and $C(e_{\text{diag}}) = 1$, so $\lambda_{sw}(e_{\text{diag}})$ is indeterminate $(0/0)$. For the cost families studied here, the denominator $\sqrt{n} - AC > 0$ for all $e \neq e_{\text{diag}}$, so $\lambda_{sw}(e) = \lambda_0$ is well-defined and $\lambda_0 = \inf_e \lambda_{sw}(e)$. For all other directions, $\lambda_{sw}(e) > 0$ is the penalty weight at which e becomes competitive with the diagonal.

Remark 1.3 (Staircase connection). For $n = 2$ and pairing cost, $d_{\text{diag}}(\lambda) = \sqrt{2}(1 + \lambda)$. The factor $\sqrt{2}$ is the staircase paradox ratio $L^1/L^2 = \sqrt{2}$ in the positive quadrant. The threshold $\lambda_0 = \frac{1}{3}$ is the unique λ at which a penalised direction first matches this diagonal cost: $d^{*2} - d_{\text{diag}}^2 = -2(3\lambda - 1)^2$. This is an observation within the $n = 2$ model.

2 Even Dimensions: Pairing Cost

2.1 Three-Phase Structure

Theorem A — Universal threshold in even dimensions

Let $C(e) = 2\sum_{i=1}^{k} e_{2i-1}e_{2i}$ (pairing cost), $n = 2k \geq 2$. Then $\lambda_0 = \frac{1}{3}$, independent of k. The phase equation factors as $d^{*2}(\lambda) - d_{\text{diag}}^2(\lambda) = -2k(3\lambda - 1)^2$.

Proof. On the equality set $e_{2i-1} = e_{2i} = s_i/2$ (AM-GM optimum), $C = 2\sum_i (s_i/2)^2 = \sum_i s_i^2/2$. The constraint $|e|^2 = 1$ gives $\sum_i s_i^2 = 2$. At the optimum (all s_i equal, $s_i = p/k$): $C_{\text{opt}} = p^2/(2k)$. After normalising $C_{\text{opt}}(p_{\text{diag}}) = 1$ (where $p_{\text{diag}} = \sqrt{2k}$), the k-dependence cancels. Substituting into $d(e, \lambda) = (2/p)(1 + \lambda C_{\text{opt}})$ and optimising over p:

$$d^{*2} - d_{\text{diag}}^2 = -2k(3\lambda - 1)^2,$$

giving the double root $\lambda_0 = \frac{1}{3}$. $\square$

2.2 Block-Repetition Inheritance

Theorem B — Block-repetition identity

For $n = 2k$ and $u = (u_1, u_2) \in S_+^1$, let $\hat{e}_{2k}(u) = (u_1, u_2, \ldots, u_1, u_2)/\sqrt{k} \in S_+^{2k-1}$. Then $\lambda_{sw,2k}(\hat{e}_{2k}(u)) = \lambda_{sw,2}(u)$ (exact identity).

Proof. By degree-2 homogeneity, $C_{2k}(\hat{e}) = C_2(u)$. The sum $\sum \hat{e}_i = \sqrt{k}\sum u_i$, so $A_{2k}(\hat{e}) = \sqrt{k}A_2(u)$. In $\lambda_{sw} = (A - \sqrt{n})/(\sqrt{n} - AC)$: numerator $= \sqrt{k}(A_2 - \sqrt{2})$, denominator $= \sqrt{k}(\sqrt{2} - A_2C_2)$; $\sqrt{k}$ cancels. □

Squeezing: Theorem A gives $\inf_e \lambda_{sw,2k}(e) \geq \frac{1}{3}$ (independent lower bound). The block-repeated family $\hat{e}_{2k}(e_K)$ with $e_K = (K, K-1)/\|e_K\|$ satisfies $\lambda_{sw,2k}(\hat{e}_{2k}(e_K)) \to \frac{1}{3}$ as $K \to \infty$. Squeezing gives $\lambda_0(2k) = \frac{1}{3}$.

Three necessary conditions: degree-2 homogeneity of C; pairing (block-diagonal) structure; n even. For odd n, no integer k exists; the mechanism fails structurally.

3 Odd Dimensions: Cyclic Cost

3.1 Global Minimum for $n = 3$

Theorem C — $n = 3$ global minimum

For the cyclic cost $C(e) = e_1e_2 + e_2e_3 + e_3e_1$ in $n = 3$, $\lambda_{sw}(e) \geq \frac{1}{2}$ for all $e \in S^2_+$, with equality iff $e \to e_{\text{diag}}$. Hence $\lambda_0 = \frac{1}{2}$.

Proof. Set $s = e_1 + e_2 + e_3$ and $q = e_1e_2 + e_2e_3 + e_3e_1$. Since $|e|^2 = s^2 - 2q = 1$, we have $q = (s^2 - 1)/2$. The condition $\lambda_{sw} \geq \frac{1}{2}$ is equivalent to $A(2 + C) \geq 3\sqrt{3}$:

$$\frac{3(s^2 + 3)}{2s} \geq 3\sqrt{3} \iff s^2 - 2\sqrt{3}\,s + 3 = (s - \sqrt{3})^2 \geq 0.$$

□

Remark 3.1. $n = 3$ is the unique dimension where q is fully determined by s via $q = (s^2 - 1)/2$ (three-variable symmetric polynomial identity), collapsing the problem to a perfect square.

3.2 General Odd Dimensions

Theorem D — Exact formula for all odd n

For cyclic cost in odd dimension $n \geq 3$:

$$\lambda_0(n) = \frac{1}{1 + 2\cos(\pi/n)}.$$

Proof (five steps). Write $e = \sum_k \alpha_k v_k$ in the cyclic eigenbasis (eigenvalues $\mu_k = 2\cos(2\pi k/n)$, $v_0 = e_{\text{diag}}$). Set $a = \alpha_0 = p/\sqrt{n} \in (0, 1]$. The normalised cyclic cost satisfies $C = C_{\text{raw}}/2$ where $C_{\text{raw}} = \sum_k \mu_k|\alpha_k|^2$. The switch value simplifies to $\lambda_{sw} = (1 - a)/(a - C_{\text{raw}}/2)$.

Step 1. Since $\partial\lambda_{sw}/\partial C_{\text{raw}} < 0$, minimising λ_{sw} requires minimising C_{raw}. With $\sum_{k\geq 1} |\alpha_k|^2 = 1 - a^2$, the minimum is $C_{\text{raw}} \geq 2a^2 + \lambda_{\min}(1 - a^2)$ where $\lambda_{\min} = -2\cos(\pi/n)$ (most-negative eigenvalue at $k_{\min} = (n - 1)/2$).

Step 2. Substituting: $\lambda_{sw}(e) \geq \lambda^{\min}_{sw}(a) = 1/(a + \cos(\pi/n)(1 + a))$.

Step 3. $d\lambda^{\min}_{sw}/da = -(1 + \cos\pi/n)/(\cdots)^2 < 0$ (since $\cos(\pi/n) > -1$ for all $n \geq 2$).

Step 4. $\inf_{a\in(0,1]} \lambda^{\min}_{sw}(a) = \lim_{a\to 1^-} \lambda^{\min}_{sw}(a) = 1/(1 + 2\cos\pi/n)$. (Verified symbolically: difference is identically zero.)

Step 5. The $k_{\min}$-mode direction realises this infimum as $\beta \to 0^+$. □

3.3 Algebraic Bifurcation at $n = 3$

Theorem E — Bifurcation formula

Let $e(t) = (1+t, 1, \ldots, 1, 1-(n-1)t)/\|\cdot\|$ for $t \to 0^+$. Then

$$L(n) := \lim_{t\to 0^+} \lambda_{sw}(e(t)) = \frac{(n-1)(n^2-2n+4)}{n^3+n^2-4n+4},$$

and

$$L(n) - \frac{n-1}{n+1} = \frac{-2n(n-3)(n-1)}{(n+1)(n^3+n^2-4n+4)}.$$

The denominator is positive for all $n \geq 1$. Therefore: $n = 3$: $L(3) = \frac{1}{2} = (n-1)/(n+1)$ (diagonal locking, factor $n-3=0$). Odd $n \geq 5$: $L(n) < (n-1)/(n+1)$ (diagonal optimality fails).

Proof. Direct polynomial computation. □

Interpretation. $n = 3$ is the exact algebraic bifurcation point: the asymmetric perturbation branch is tangent to the diagonal limit at $n = 3$ and strictly beats it for all odd $n \geq 5$.

4 All Dimensions: Full-Connection Cost and A_{n-1} Lattices

Theorem F — Full-connection cost

For $C_{\text{full}}(e) = \sum_{i<j} e_i e_j$ in any dimension $n \geq 2$:

$$d^{*2}(\lambda) - d^2_{\text{diag}}(\lambda) = -\frac{n\big(\lambda(n+1)-(n-1)\big)^2}{(n-1)^2}, \qquad \lambda_0 = \frac{n-1}{n+1}.$$

The equality set is $\{|e| = 1, \sum e_i = p\} = S^{n-2}$, and C_{full} is the natural cost of the A_{n-1} root lattice.

Proof. Since $C_{\text{full}} = (\sum e_i)^2 - |e|^2)/2 = (p^2-1)/2$ depends only on $p = \sum e_i$, the equality set is S^{n-2}_p for each p. Normalising gives $\alpha_n = n-1$; substituting into $d^*(\lambda) = c_n\sqrt{\lambda(\alpha_n - \lambda)}$ yields the stated double root. □

Lattice correspondence: $C_{\text{full}} = 1 - N_{A_{n-1}} \cdot (n-1)^{-1}$ where $N_{A_{n-1}} = 1 - p^2/n$ is the A_{n-1} lattice norm on the unit sphere. The cyclic cost coincides with C_{full} only for $n = 2, 3$ (where every pair is adjacent), explaining why the A_1/A_2 cases were exact.

5 Asymptotic Behaviour as $n \to \infty$

Proposition 5.1 (Asymptotic expansion)**.** *As $n \to \infty$ (odd), the cyclic threshold satisfies*

$$\lambda_0(n) = \frac{1}{1+2\cos(\pi/n)} = \frac{1}{3} + \frac{\pi^2}{9n^2} + \frac{\pi^4}{36n^4} + O(n^{-6}).$$

Cost family	Dimensions	λ_0	Limit $n \to \infty$
Pairing	Even $n = 2k$	$1/3$ (exact)	$1/3$
Cyclic	Odd n	$1/(1+2\cos\pi/n)$	$1/3$, rate $\pi^2/(9n^2)$
Full-connection	All n	$(n-1)/(n+1)$	1, rate $2/n$

6 Asymmetric Targets

We now consider a general target $r = (r_1, \ldots, r_n)$ with $r_i > 0$, setting $A_r(e) = n/\sum_i (e_i/r_i)$ and $d^r_{\text{diag}}(\lambda) = |r|(1+\lambda)$.

Theorem G — Asymmetric target dichotomy

Let $r = (r_1, \ldots, r_n)$ with $r_i > 0$.
(i) Symmetric case ($r_1 = \cdots = r_n$): reduces to the symmetric theory; λ_0 is as in Theorems A–F.
(ii) Asymmetric case (r_i not all equal): the direction $e^* = (1/r_1, \ldots, 1/r_n)/|e^*|$ satisfies

$$d_r(e^*, \lambda) < d^r_{\text{diag}}(\lambda) \quad \text{for all } \lambda \geq 0.$$

The diagonal is never optimal; $\lambda_0 = 0$ and the threshold is degenerate.

Proof. Set $f_i = e_i/r_i$. On the ellipsoid $\sum r_i^2 f_i^2 = 1$, Cauchy–Schwarz gives

$$\sum_i f_i = \sum_i (r_i f_i) \cdot \frac{1}{r_i} \leq \sqrt{\sum_i r_i^2 f_i^2} \cdot \sqrt{\sum_i \frac{1}{r_i^2}} = |r^{-1}|,$$

with equality iff $r_i f_i \propto 1/r_i$, i.e. $e \propto (1/r_1, \ldots, 1/r_n) = e^*$. Thus $A_r(e^*) = n/|r^{-1}|$.

By Cauchy–Schwarz applied to $(r_1, \ldots, r_n)$ and $(1/r_1, \ldots, 1/r_n)$:

$$n = \sum_i r_i \cdot \frac{1}{r_i} \leq |r| \cdot |r^{-1}|,$$

with equality iff $r_1 = \cdots = r_n$. Hence when r is not constant, $A_r(e^*) = n/|r^{-1}| < |r|$.

Since $A_r(e^*) < |r| = d^r_{\text{diag}}(0)/1$, the numerator of $\lambda_{sw}(e^*)$ is negative, so $d_r(e^*, \lambda) < d^r_{\text{diag}}(\lambda)$ for all $\lambda \geq 0$. □

Corollary 6.1 (Symmetry is necessary and sufficient)**.** *$\lambda_0 > 0$ if and only if $r_1 = \cdots = r_n$. Equivalently: $|r| \cdot |r^{-1}| = n$ iff r is constant (Cauchy–Schwarz equality).*

7 A_2 Naturality and Theorem N

7.1 A_2 Naturality

Theorem H — A_2 naturality

Among all S_3-symmetric normalised quadratic costs on S^2_+:

1. *Unique decomposition:* $C = \alpha + (1-\alpha)C_{\text{cyclic}}$, $\alpha \in [0, 1)$. (Proof: S_3-invariant quadratics have basis $\{|e|^2, C_{\text{cyclic}}\}$; normalisation gives $a + b = 1$.)
2. *Strict monotonicity:* $\lambda_0(\alpha)$ strictly increasing in α.
3. *Extremality:* C_{cyclic} ($\alpha = 0$) uniquely minimises λ_0.
4. *Lattice origin:* $C_{\text{cyclic}} = 1 - N_{A_2}$ where $N_{A_2} = \frac{1}{2}\sum_{i<j}(e_i - e_j)^2$ is the A_2 norm.

Boundary: adding cubic $q = e_1 e_2 e_3$ achieves $\lambda_0 < \frac{1}{2}$; the quadratic class is minimal for C_{cyclic} to be extremal.

7.2 Theorem N and Its Extension

Theorem N — Global minimiser condition (extended)

Finite case: For Type IV cost with finite Fourier expansion and $\alpha[C] > 0$: $\lambda_0 = \min_\theta \lambda_{sw}(\theta)$.
Infinite case (Theorem N'): If $|a_k| \le Me^{-\sigma k}$ ($\sigma > 0$) and $\alpha[C] > 0$ (equivalently $C_2 < -1$), then: (1) $\lambda_0 = \min_\theta \lambda_{sw}(\theta)$; (2) $|R_M(K)| \le C_\sigma^{M+1} e^{-2\sigma(M+1)K}/(1 - C_\sigma e^{-2\sigma K})$.
Key lemma: $h(\varphi) = C(\pi/4 + \varphi) - \cos\varphi$ satisfies $\mathrm{Re}(h(i\tau)) = -(2C_2 + 1)\tau^2/2 + O(\tau^4) > 0$ for small $\tau > 0$, ensuring the denominator is nonzero in a strip $|\mathrm{Im}(\varphi)| < \delta$.

8 Spectral Unification

The three cost families—pairing, cyclic, full-connection—admit a single formula for λ_0 in terms of the spectrum of M.

Theorem S — Spectral unification formula

Let $C = e^T M e$ be a normalised quadratic cost with $Me_{\mathrm{diag}} = e_{\mathrm{diag}}$. Define

$$\lambda_{\min} \;=\; \min \mathrm{spec}\big(M\big|_{e_{\mathrm{diag}}^\perp}\big),$$

the smallest eigenvalue of M restricted to the hyperplane $\{v : \langle v, e_{\mathrm{diag}}\rangle = 0\}$. Then

$$\lambda_0 \;=\; \frac{1}{1 - 2\lambda_{\min}}.$$

Proof. Write $e = se_{\mathrm{diag}} + \sqrt{1-s^2}\, v$ with $v \perp e_{\mathrm{diag}}$, $|v| = 1$, $s = \langle e, e_{\mathrm{diag}}\rangle \in (0, 1]$.
Step 1 (expansions). Since $Me_{\mathrm{diag}} = e_{\mathrm{diag}}$ and $v \perp e_{\mathrm{diag}}$:

$$A(e) = \frac{n}{\sum e_i} = \frac{\sqrt{n}}{s}, \qquad C(e) = e^T M e = s^2 + (1 - s^2)(v^T M v).$$

The v-dependent quantity $\mu := v^T M v$ satisfies $\mu \ge \lambda_{\min}$ by definition of $\lambda_{\min}$.

Step 2 (reduction to sign check). The condition $\lambda_{sw}(e) \ge \lambda_0$ is equivalent to $d(e, \lambda_0) \ge d_{\mathrm{diag}}(\lambda_0)$, i.e. $A(e)(1 + \lambda_0 C(e)) \ge \sqrt{n}(1 + \lambda_0)$. Dividing by $\sqrt{n}(1 + \lambda_0)$ and using $A(e) = \sqrt{n}/s$:

$$f(s, \mu) := \frac{1 + \lambda_0(s^2 + (1 - s^2)\mu)}{s(1 + \lambda_0)} - 1 \;\ge\; 0.$$

Algebraic simplification gives

$$f(s, \mu) = \frac{1 - s}{s} \cdot g(s, \mu), \qquad g(s, \mu) := 1 - \lambda_0 s + \lambda_0(1 + s)\mu.$$

Step 3 (sign of g). Since $\partial g/\partial\mu = \lambda_0(1 + s) > 0$, the worst case is $\mu = \lambda_{\min}$. Substituting $\lambda_0 = 1/(1 - 2\lambda_{\min})$:

$$g(s, \lambda_{\min}) = \frac{(1 - \lambda_{\min})(s - 1)}{2\lambda_{\min} - 1}.$$

When $\lambda_{\min} < 0$ (required for $\lambda_0 < 1$): $(1 - \lambda_{\min}) > 0$, $(s - 1) \le 0$, $(2\lambda_{\min} - 1) < 0$. Hence $g \ge 0$, with equality iff $s = 1$.

Conclusion. $(1 - s)/s \ge 0$ and $g \ge 0$, so $f \ge 0$ for all $e \in S_+^{n-1}$. Equality holds iff $s = 1$, i.e. $e = e_{\mathrm{diag}}$. Hence $\lambda_{sw}(e) \ge \lambda_0$ for all e, with the infimum approached as $e \to e_{\mathrm{diag}}$ along the $\lambda_{\min}$-eigenvector direction. □

Corollary 8.1 (No counterexamples exist)**.** *For any normalised quadratic cost $C = e^T M e$ with $Me_{\text{diag}} = e_{\text{diag}}$ and $\lambda_{\min}(M|_{e_{\text{diag}}^\perp}) < 0$, the formula $\lambda_0 = 1/(1 - 2\lambda_{\min})$ holds exactly. There is no quadratic cost satisfying the normalisation for which the formula fails.*

Verification across all cost families.

Cost	n	$\lambda_{\min}(M\|_\perp)$	$1/(1-2\lambda_{\min})$	True λ_0
Pairing	$2k$	-1	$1/3$	$1/3$
Cyclic	3	$-1/2$	$1/2$	$1/2$
Cyclic	5	$-(\sqrt{5}-1)/2$	$(3-\sqrt{5})/2$	$(3-\sqrt{5})/2$
Cyclic	odd n	$-\cos(\pi/n)$	$1/(1+2\cos\pi/n)$	$1/(1+2\cos\pi/n)$
Full-conn.	n	$-1/(n-1)$	$(n-1)/(n+1)$	$(n-1)/(n+1)$

Machine-verified for 14 cases (zero failures).

Why this works. The key is that $A_2 = \sqrt{n}/2$ is *universal*—it depends only on the geometry of the sphere, not on C. The entire cost-dependence of λ_0 is therefore carried by C_2, which equals $\lambda_{\min}(M|_\perp) - 1$. The formula $\lambda_0 = 1/(1 - 2\lambda_{\min})$ is the direct translation of this into the threshold.

Relation to earlier theorems. Theorem A ($\lambda_0 = \frac{1}{3}$): pairing cost has $\lambda_{\min} = -1$ for all k. Theorem D ($\lambda_0 = 1/(1 + 2\cos\pi/n)$): cyclic cost has $\lambda_{\min} = -\cos(\pi/n)$. Theorem F ($\lambda_0 = (n-1)/(n+1)$): full-connection cost has $\lambda_{\min} = -1/(n-1)$. All three are instances of $\lambda_0 = 1/(1 - 2\lambda_{\min})$.

Scope of Theorem S. The proof requires: (i) $C = e^T M e$ quadratic; (ii) $Me_{\text{diag}} = e_{\text{diag}}$ (normalisation); (iii) $\lambda_{\min}(M|_{e_{\text{diag}}^\perp}) < 0$ (so that $\lambda_0 < 1$). Under these three conditions the argument is a complete global lower bound—the Corollary shows no counterexamples can exist. All three cost families are direct instances; no additional assumptions are needed.

9 Connections to Spectral Theory and Optimisation

Theorem S places the optimality threshold in contact with three classical frameworks.

9.1 Rayleigh Quotient and Courant–Fischer

The quantity $\lambda_{\min}(M|_{e_{\text{diag}}^\perp})$ is the *constrained Rayleigh quotient minimum*:

$$\lambda_{\min}(M|_{e_{\text{diag}}^\perp}) = \min_{\substack{v \perp e_{\text{diag}} \\ |v|=1}} v^T M v,$$

which equals the smallest eigenvalue of M in the hyperplane orthogonal to e_{diag}, given by Courant–Fischer. Theorem S is the statement that λ_0 is a strictly decreasing function of this Rayleigh minimum:

$$\lambda_0 = \frac{1}{1 - 2\lambda_{\min}}, \qquad \frac{d\lambda_0}{d\lambda_{\min}} = \frac{2}{(1 - 2\lambda_{\min})^2} > 0.$$

The more negative the most unstable direction, the smaller λ_0—the diagonal is easier to displace. Conversely, $\lambda_{\min} \to 0^-$ gives $\lambda_0 \to \infty$: the diagonal is never displaced.

9.2 Diagonal Stability and Lyapunov Analogy

Define the *excess cost*

$$L_\lambda(e) := d(e,\lambda) - d_{\text{diag}}(\lambda) = A(e)(1+\lambda C(e)) - \sqrt{n}(1+\lambda).$$

This function satisfies $L_\lambda(e_{\text{diag}}) = 0$ and

$$L_\lambda(e) \geq 0 \ \forall e \iff \lambda \leq \lambda_0 \iff \text{diagonal is globally optimal at penalty } \lambda.$$

The threshold λ_0 is precisely the parameter at which L_λ loses non-negativity—the *stability boundary* of the diagonal direction. This mirrors Lyapunov stability theory, where a function $V(x) \geq 0$ certifies stability and the boundary $V = 0$ marks the critical parameter. Here L_λ plays the role of V, and λ_0 is the critical penalty at which diagonal optimality fails.

9.3 Linearisation via Fractional Parametrisation

The parametrisation $e = se_{\text{diag}} + \sqrt{1-s^2}v$ reduces λ_{sw} to a fractional linear form:

$$\lambda_{sw}(s,\mu) = \frac{1}{s-(1+s)\mu}, \qquad s \in (0,1], \quad \mu = v^T M v \in [\lambda_{\min}, 1].$$

The optimisation $\lambda_0 = \inf_{s,\mu} \lambda_{sw}(s,\mu)$ transforms to

$$\lambda_0 = \frac{1}{\max\limits_{\substack{s\in(0,1]\\ \mu\in[\lambda_{\min},1]}} \big(s-(1+s)\mu\big)}.$$

The objective $h(s,\mu) = s-(1+s)\mu$ is *linear* in both s and μ. Its maximum over the rectangle $[0,1]\times[\lambda_{\min},1]$ is attained at the corner $(s^*,\mu^*) = (1,\lambda_{\min})$, giving $h_{\max} = 1-2\lambda_{\min}$, and hence $\lambda_0 = 1/(1-2\lambda_{\min})$.

This converts the original non-linear sphere optimisation into a two-variable linear programme whose vertex solution recovers the spectral formula directly.

Remark 9.1. The three frameworks are consistent: Courant–Fischer identifies $\lambda_{\min}$; the LP corner solution at $\mu = \lambda_{\min}$, $s = 1$ matches the Rayleigh minimiser; and $L_\lambda \geq 0$ is the certificate of diagonal stability up to λ_0.

10 Complete Results Table

Result	Status	Method	Section
Theorem A: $\lambda_0 = \frac{1}{3}$, even $n = 2k$	Exact	Double root $-2k(3\lambda-1)^2$	§2.1
Theorem B: block-repetition identity	Exact	Degree-2 homogeneity	§2.2
Theorem C: $n = 3$ global min, $(s-\sqrt{3})^2 \geq 0$	Exact	Perfect square	§3.1
Theorem D: $\lambda_0 = 1/(1+2\cos\pi/n)$, odd n	Exact	Fourier decomp. + monotonicity	§3.2
Theorem E: bifurcation factor $n-3$	Exact	Polynomial factorisation	§3.3
Theorem F: full-connection $\lambda_0 = (n-1)/(n+1)$	Exact	Double root, $\alpha_n = n-1$	§4
Asymptotic $\lambda_0 = \frac{1}{3} + \pi^2/(9n^2) + \cdots$	Exact	Series expansion	§5
Theorem G: asymmetric target dichotomy	Exact	Cauchy–Schwarz	§6
Theorem H: A_2 naturality	Exact	Representation theory	§7.1
Theorem N': analytic extension	Exact	Complex analysis	§7.2
Theorem S: $\lambda_0 = 1/(1-2\lambda_{\min}(M\|_\perp))$	Exact	Spectral + global lower bound	§8
LP linearisation of $\inf \lambda_{sw}$	Exact	Fractional parametrisation	§9.3

11 Scope and Open Problems

11.1 What Is and Is Not Claimed

The term "phase transition" is used loosely for "optimality threshold"; there is no probabilistic structure. Results depend on the chosen cost families; different choices yield different thresholds. Within each fixed cost family, the threshold is determined by the algebraic structure of that cost—not independently tunable.

11.2 Open Problems

1. **Asymmetric targets: positive threshold case.** When r_i are not all equal, Theorem G shows $\lambda_0 = 0$. A natural follow-up: for which cost families C_r adapted to r does a positive threshold re-emerge?

2. **Algebraic connection to k^2.** The asymptotic $\lambda_{sw}(K) - \frac{1}{3} \sim \alpha[C]/K^2$ links $\frac{1}{3}$ and K^{-2} through the same expansion. Whether this connects to the counting formula $|W(m,h)| = A(m,h)^2$ from [1] remains open.

3. **Non-cyclic/non-pairing costs in odd dimensions.** The classification Type E/T/F covers pairing and cyclic costs. Other cost families in odd dimensions are not characterised.

All theorems proved without numerical input. Machine verification: 25+14 assertions (SymPy/mpmath ≥ 40 digits; NumPy), zero failures. No conjectures remain. The central result (Theorem S) unifies all cost families via a single spectral formula, proved as a global lower bound and independently derived via linear programming.

References

[1] Xie, K. *KFX-R: Rogers–Ramanujan Truncations in Step-Count Layers*. Zenodo, 2026. https://doi.org/10.5281/zenodo.19185663